

```
SP500: 21
● @NicolasPiquet →/workspaces/esilv-markets-pulse-g01-HANETeam (Nicolas_Piquet) $ grep -c AAPL data/sample/prices.csv
grep -c SP500 data/sample/prices.csv
21
21
○ @NicolasPiquet →/workspaces/esilv-markets-pulse-g01-HANETeam (Nicolas_Piquet) $
```

```
● @NicolasPiquet →/workspaces/esilv-markets-pulse-g01-HANETeam (Nicolas_Piquet) $ python --version
git --version
python src/main.py
Python 3.14.2
git version 2.55.0
=== MarketPulse ===
```

```
Instrument
AAPL - Apple Inc.
Last price: 266.20 USD
```

```
Benchmark
SP500 - S&P 500
Last level: 6742.00
```

```
Period: 1 month
Interval: Daily
```

```
Observations
AAPL: 21
SP500: 21
```

```
○ @NicolasPiquet →/workspaces/esilv-markets-pulse-g01-HANETeam (Nicolas_Piquet) $
```

```
● @NicolasPiquet →/workspaces/esilv-markets-pulse-g01-HANETeam (Nicolas_Piquet) $ grep AAPL data/sample/prices.csv
grep SP500 data/sample/prices.csv
2026-09-01,AAPL,249.20,251.50,248.00,250.00,38000000
2026-09-02,AAPL,250.80,252.80,249.60,251.30,39500000
2026-09-03,AAPL,249.60,251.30,248.40,249.80,41000000
2026-09-04,AAPL,251.30,253.60,250.10,252.10,42500000
2026-09-08,AAPL,251.90,253.90,250.70,252.40,44000000
2026-09-09,AAPL,254.00,255.70,252.80,254.20,38000000
2026-09-10,AAPL,254.30,256.60,253.10,255.10,39500000
2026-09-11,AAPL,254.20,256.20,253.00,254.70,41000000
2026-09-14,AAPL,256.60,258.30,255.40,256.80,42500000
2026-09-15,AAPL,257.20,259.50,256.00,258.00,44000000
2026-09-16,AAPL,256.70,258.70,255.50,257.20,38000000
2026-09-17,AAPL,259.20,260.90,258.00,259.40,39500000
2026-09-18,AAPL,259.30,261.60,258.10,260.10,41000000
2026-09-21,AAPL,259.10,261.10,257.90,259.60,42500000
2026-09-22,AAPL,261.10,262.80,259.90,261.30,44000000
2026-09-23,AAPL,261.20,263.50,260.00,262.00,38000000
2026-09-24,AAPL,263.00,265.00,261.80,263.50,39500000
2026-09-25,AAPL,262.60,264.30,261.40,262.80,41000000
2026-09-28,AAPL,263.30,265.60,262.10,264.10,42500000
2026-09-29,AAPL,264.50,266.50,263.30,265.00,44000000
2026-09-30,AAPL,266.00,267.70,264.80,266.20,38000000
2026-09-01,SP500,6592.00,6615.00,6580.00,6600.00,0
2026-09-02,SP500,6607.00,6627.00,6595.00,6612.00,0
2026-09-03,SP500,6596.00,6613.00,6584.00,6598.00,0
2026-09-04,SP500,6612.00,6635.00,6600.00,6620.00,0
2026-09-08,SP500,6623.00,6643.00,6611.00,6628.00,0
2026-09-09,SP500,6640.00,6657.00,6628.00,6642.00,0
2026-09-10,SP500,6642.00,6665.00,6630.00,6650.00,0
2026-09-11,SP500,6642.00,6662.00,6630.00,6647.00,0
2026-09-14,SP500,6663.00,6680.00,6651.00,6665.00,0
2026-09-15,SP500,6664.00,6687.00,6652.00,6672.00,0
```

```

}
● @NicolasPiquet → /workspaces/esilv-marketpulse-g01-HANETeam (Nicolas_Piquet) $ head data/sample/prices.csv
date,ticker,open,high,low,close,volume
2026-09-01,AAPL,249.20,251.50,248.00,250.00,38000000
2026-09-01,SP500,6592.00,6615.00,6580.00,6600.00,0
2026-09-02,AAPL,250.80,252.80,249.60,251.30,39500000
2026-09-02,SP500,6607.00,6627.00,6595.00,6612.00,0
2026-09-03,AAPL,249.60,251.30,248.40,249.80,41000000
2026-09-03,SP500,6596.00,6613.00,6584.00,6598.00,0
2026-09-04,AAPL,251.30,253.60,250.10,252.10,42500000
2026-09-04,SP500,6612.00,6635.00,6600.00,6620.00,0
2026-09-08,AAPL,251.90,253.90,250.70,252.40,44000000
○ @NicolasPiquet → /workspaces/esilv-marketpulse-g01-HANETeam (Nicolas_Piquet) $ 
bloomberg_reference_expected.json bloomberg_reference_sample.json instruments.json prices.csv
● @NicolasPiquet → /workspaces/esilv-marketpulse-g01-HANETeam/data/sample (Nicolas_Piquet) $ cd ../../
pwd
/workspaces/esilv-marketpulse-g01-HANETeam
● @NicolasPiquet → /workspaces/esilv-marketpulse-g01-HANETeam (Nicolas_Piquet) $ cat data/sample/instruments.json
{
  "instrument": {
    "ticker": "AAPL",
    "name": "Apple Inc.",
    "currency": "USD",
    "market": "NASDAQ"
  },
  "benchmark": {
    "ticker": "SP500",
    "name": "S&P 500",
    "currency": "USD",
    "market": "US"
  }
}
○ @NicolasPiquet → /workspaces/esilv-marketpulse-g01-HANETeam (Nicolas_Piquet) $ █
@NicolasPiquet → /workspaces/esilv-marketpulse-g01-HANETeam (Nicolas_Piquet) $ cd data
pwd
ls

cd sample
pwd
ls
/workspaces/esilv-marketpulse-g01-HANETeam/data
sample
/workspaces/esilv-marketpulse-g01-HANETeam/data/sample
bloomberg_reference_expected.json bloomberg_reference_sample.json instruments.json prices.csv
○ @NicolasPiquet → /workspaces/esilv-marketpulse-g01-HANETeam/data/sample (Nicolas_Piquet) $ █

```

```

● @NicolasPiquet → /workspaces/esilv-marketpulse-g01-HANETeam (Nicolas_Piquet) $ pwd
ls
ls -la
/workspaces/esilv-marketpulse-g01-HANETeam
CONTRIBUTING.md README.md TEAM.md config data evidence labs readiness requirements.txt src
total 64
drwxrwxrwx+ 9 codespace root 4096 Oct 7 13:43 .
drwxr-xrwx+ 5 codespace root 4096 Oct 7 13:43 ..
drwxrwxrwx+ 8 codespace root 4096 Oct 7 13:44 .git
-rw-rw-rw- 1 codespace root 178 Oct 7 13:43 .gitignore
-rw-rw-rw- 1 codespace root 2943 Oct 7 13:43 CONTRIBUTING.md
-rw-rw-rw- 1 codespace root 10270 Oct 7 13:43 README.md
-rw-rw-rw- 1 codespace root 1497 Oct 7 13:43 TEAM.md
drwxrwxrwx+ 2 codespace root 4096 Oct 7 13:43 config
drwxrwxrwx+ 3 codespace root 4096 Oct 7 13:43 data
drwxrwxrwx+ 2 codespace root 4096 Oct 7 13:43 evidence
drwxrwxrwx+ 2 codespace root 4096 Oct 7 13:43 labs
drwxrwxrwx+ 2 codespace root 4096 Oct 7 13:43 readiness
-rw-rw-rw- 1 codespace root 133 Oct 7 13:43 requirements.txt
drwxrwxrwx+ 2 codespace root 4096 Oct 7 13:43 src
○ @NicolasPiquet → /workspaces/esilv-marketpulse-g01-HANETeam (Nicolas_Piquet) $ █

~/config/setting.yml: secret: null
● @NicolasPiquet → /workspaces/esilv-marketpulse-g01-HANETeam (Nicolas_Piquet) $ git status
git log --oneline
On branch Nicolas_Piquet
Your branch is up to date with 'origin/Nicolas_Piquet'.

nothing to commit, working tree clean
b46aa2e (HEAD -> Nicolas_Piquet, origin/Nicolas_Piquet) Add Nicolas Piquet to the team list
44c08d8 Update team member details in TEAM.md
383ece3 (origin/elias_bejjani) Rename TEAM_TEMPLATE.md to TEAM.md
9978780 Update TEAM_TEMPLATE.md
5207db8 (origin/Adli_BOUZGUENDA) first td session
ba5c9f4 Update Readme
9fd6c8a Update Readme
bfb94a6 init projecttemplate
○ @NicolasPiquet → /workspaces/esilv-marketpulse-g01-HANETeam (Nicolas_Piquet) $ █

~/config/setting.yml: secret: null
● @NicolasPiquet → /workspaces/esilv-marketpulse-g01-HANETeam (Nicolas_Piquet) $ git status
git log --oneline
On branch Nicolas_Piquet
Your branch is up to date with 'origin/Nicolas_Piquet'.

nothing to commit, working tree clean
b46aa2e (HEAD -> Nicolas_Piquet, origin/Nicolas_Piquet) Add Nicolas Piquet to the team list
44c08d8 Update team member details in TEAM.md
383ece3 (origin/elias_bejjani) Rename TEAM_TEMPLATE.md to TEAM.md
9978780 Update TEAM_TEMPLATE.md
5207db8 (origin/Adli_BOUZGUENDA) first td session
ba5c9f4 Update Readme
9fd6c8a Update Readme
bfb94a6 init projecttemplate
○ @NicolasPiquet → /workspaces/esilv-marketpulse-g01-HANETeam (Nicolas_Piquet) $ █

```

```

@NicolasPiquet →/workspaces/esilv-marketpulse-g01-HANETeam (Nicolas_Piquet) $ find . -type f
grep -R "AAPL" .
./git/logs/refs/remotes/origin/elias_bejjani
./git/logs/HEAD
./git/info/exclude
./git/FETCH_HEAD
./README.md:Instrument : AAPL - Apple Inc.
./README.md:AAPL
./readiness/01_R8_1_STUDENT_BOOTSTRAP_AND_CLONE_VALIDATION.md:AAPL - Apple Inc.
./readiness/01_R8_1_STUDENT_BOOTSTRAP_AND_CLONE_VALIDATION.md:AAPL: 21
./readiness/01_R8_1_STUDENT_BOOTSTRAP_AND_CLONE_VALIDATION.md:grep -c AAPL data/sample/prices.csv
./readiness/01_R8_1_STUDENT_BOOTSTRAP_AND_CLONE_VALIDATION.md:AAPL
./readiness/01_R8_1_STUDENT_BOOTSTRAP_AND_CLONE_VALIDATION.md:[x] AAPL observation count is 21
./readiness/01_R8_1_STUDENT_BOOTSTRAP_AND_CLONE_VALIDATION.md:grep AAPL data/sample/prices.csv
./readiness/01_R8_1_STUDENT_BOOTSTRAP_AND_CLONE_VALIDATION.md:AAPL
./labs/TD02_PYTHON_CSV_JSON.md:Instrument : AAPL - Apple Inc.
./labs/TD02_PYTHON_CSV_JSON.md:[ ] separates AAPL and SP500 rows
./labs/TD02_PYTHON_CSV_JSON.md:AAPL -> 250.00
./labs/TD02_PYTHON_CSV_JSON.md:AAPL -> 266.20
./labs/TD02_PYTHON_CSV_JSON.md:Do not duplicate one display implementation for AAPL and another for
SP500.
./labs/TD02_PYTHON_CSV_JSON.md:AAPL - Apple Inc.
./labs/TD02_PYTHON_CSV_JSON.md:AAPL - Apple Inc.
./labs/TD02_PYTHON_CSV_JSON.md:[ ] 21 observations for AAPL
./labs/TD02_PYTHON_CSV_JSON.md:[ ] AAPL has 21 observations
./labs/TD01_BOOTSTRAP_LINUX.md:Instrument : AAPL - Apple Inc.
./labs/TD01_BOOTSTRAP_LINUX.md:grep AAPL data/sample/prices.csv
./labs/TD01_BOOTSTRAP_LINUX.md:AAPL - Apple Inc.
./labs/TD01_BOOTSTRAP_LINUX.md:AAPL: 21
./labs/TD01_BOOTSTRAP_LINUX.md:instrument : AAPL
./labs/TD01_BOOTSTRAP_LINUX.md:grep -c AAPL data/sample/prices.csv
./labs/TD01_BOOTSTRAP_LINUX.md:grep -R "AAPL" .
./TEAM.md:- Instrument: AAPL - Apple Inc.
./data/sample/bloomberg_reference_sample.json: "canonical_ticker": "AAPL",
./data/sample/bloomberg_reference_sample.json: "provider_identifier": "AAPL US Equity",
./data/sample/prices.csv:2026-09-01,AAPL,249.20,251.50,248.00,250.00,38000000

```

```

@NicolasPiquet →/workspaces/esilv-marketpulse-g01-HANETeam (Nicolas_Piquet) $ find . -type f
grep -R "AAPL" .
./git/logs/refs/remotes/origin/elias_bejjani
./git/logs/HEAD
./git/info/exclude
./git/FETCH_HEAD
./README.md:Instrument : AAPL - Apple Inc.
./README.md:AAPL
./readiness/01_R8_1_STUDENT_BOOTSTRAP_AND_CLONE_VALIDATION.md:AAPL - Apple Inc.
./readiness/01_R8_1_STUDENT_BOOTSTRAP_AND_CLONE_VALIDATION.md:AAPL: 21
./readiness/01_R8_1_STUDENT_BOOTSTRAP_AND_CLONE_VALIDATION.md:grep -c AAPL data/sample/prices.csv
./readiness/01_R8_1_STUDENT_BOOTSTRAP_AND_CLONE_VALIDATION.md:AAPL
./readiness/01_R8_1_STUDENT_BOOTSTRAP_AND_CLONE_VALIDATION.md:[x] AAPL observation count is 21
./readiness/01_R8_1_STUDENT_BOOTSTRAP_AND_CLONE_VALIDATION.md:grep AAPL data/sample/prices.csv
./readiness/01_R8_1_STUDENT_BOOTSTRAP_AND_CLONE_VALIDATION.md:AAPL
./labs/TD02_PYTHON_CSV_JSON.md:Instrument : AAPL - Apple Inc.
./labs/TD02_PYTHON_CSV_JSON.md:[ ] separates AAPL and SP500 rows
./labs/TD02_PYTHON_CSV_JSON.md:AAPL -> 250.00
./labs/TD02_PYTHON_CSV_JSON.md:AAPL -> 266.20
./labs/TD02_PYTHON_CSV_JSON.md:Do not duplicate one display implementation for AAPL and another for
SP500.
./labs/TD02_PYTHON_CSV_JSON.md:AAPL - Apple Inc.
./labs/TD02_PYTHON_CSV_JSON.md:AAPL - Apple Inc.
./labs/TD02_PYTHON_CSV_JSON.md:[ ] 21 observations for AAPL
./labs/TD02_PYTHON_CSV_JSON.md:[ ] AAPL has 21 observations
./labs/TD01_BOOTSTRAP_LINUX.md:Instrument : AAPL - Apple Inc.
./labs/TD01_BOOTSTRAP_LINUX.md:grep AAPL data/sample/prices.csv
./labs/TD01_BOOTSTRAP_LINUX.md:AAPL - Apple Inc.
./labs/TD01_BOOTSTRAP_LINUX.md:AAPL: 21
./labs/TD01_BOOTSTRAP_LINUX.md:instrument : AAPL
./labs/TD01_BOOTSTRAP_LINUX.md:grep -c AAPL data/sample/prices.csv
./labs/TD01_BOOTSTRAP_LINUX.md:grep -R "AAPL" .
./TEAM.md:- Instrument: AAPL - Apple Inc.
./data/sample/bloomberg_reference_sample.json: "canonical_ticker": "AAPL",
./data/sample/bloomberg_reference_sample.json: "provider_identifier": "AAPL US Equity",
./data/sample/prices.csv:2026-09-01,AAPL,249.20,251.50,248.00,250.00,38000000

```

```

● @NicolasPiquet → /workspaces/esilv-marketpulse-g01-HANETeam (Nicolas_Piquet) $ find . -type f
grep -R "AAPL" .
./README.md
./readiness/01_R8_1_STUDENT_BOOTSTRAP_AND_CLONE_VALIDATION.md
./evidence/README.md
./requirements.txt
./src/main.py
./labs/README.md
./labs/TD02_PYTHON_CSV_JSON.md
./labs/TD01_BOOTSTRAP_LINUX.md
./labs/TD03_GIT_LOCAL_WORKFLOW.md
./TEAM.md
./data/sample/bloomberg_reference_sample.json
./data/sample/prices.csv
./data/sample/bloomberg_reference_expected.json
./data/sample/instruments.json
./CONTRIBUTING.md
./config/settings.yml
./.gitignore
./.git/description
./.git/packed-refs
./.git/index
./.git/hooks/pre-push
./.git/hooks/push-to-checkout.sample
./.git/hooks/sendemail-validate.sample
./.git/hooks/commit-msg.sample
./.git/hooks/post-update.sample
./.git/hooks/fsmonitor-watchman.sample
./.git/hooks/pre-rebase.sample
./.git/hooks/pre-merge-commit.sample
./.git/hooks/pre-receive.sample

```

```

● @NicolasPiquet → /workspaces/esilv-marketpulse-g01-HANETeam (Nicolas_Piquet) $ ls -lh

```

```

head -n 5 data/sample/prices.csv
tail -n 5 data/sample/prices.csv
total 48K
-rw-rw-rw- 1 codespace root 2.9K Oct  7 13:43 CONTRIBUTING.md
-rw-rw-rw- 1 codespace root 11K Oct  7 13:43 README.md
-rw-rw-rw- 1 codespace root 1.5K Oct  7 13:43 TEAM.md
drwxrwxrwx+ 2 codespace root 4.0K Oct  7 13:43 config
drwxrwxrwx+ 3 codespace root 4.0K Oct  7 13:43 data
drwxrwxrwx+ 2 codespace root 4.0K Oct  7 13:43 evidence
drwxrwxrwx+ 2 codespace root 4.0K Oct  7 13:43 labs
drwxrwxrwx+ 2 codespace root 4.0K Oct  7 13:43 readiness
-rw-rw-rw- 1 codespace root 133 Oct  7 13:43 requirements.txt
drwxrwxrwx+ 2 codespace root 4.0K Oct  7 13:43 src
date,ticker,open,high,low,close,volume
2026-09-01,AAPL,249.20,251.50,248.00,250.00,38000000
2026-09-01,SP500,6592.00,6615.00,6580.00,6600.00,0
2026-09-02,AAPL,250.80,252.80,249.60,251.30,39500000
2026-09-02,SP500,6607.00,6627.00,6595.00,6612.00,0
2026-09-28,SP500,6720.00,6743.00,6708.00,6728.00,0
2026-09-29,AAPL,264.50,266.50,263.30,265.00,44000000
2026-09-29,SP500,6729.00,6749.00,6717.00,6734.00,0
2026-09-30,AAPL,266.00,267.70,264.80,266.20,38000000
2026-09-30,SP500,6740.00,6757.00,6728.00,6742.00,0

```

```

● @NicolasPiquet → /workspaces/esilv-marketpulse-g01-HANETeam (Nicolas_Piquet) $ █

```